Activity 7

The damped harmonic oscillator in 1D is subject to $F_x^{\text{spring}} = -kx$ and $F_x^{\text{drag}} = -b\dot{x}$ lead to the following EOM:

$$m\ddot{x} + b\dot{x} + kx = 0\,. \tag{1}$$

Using a trial solution $x(t) = e^{rt}$, we found the general solution to the EOM to be

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}\,, \qquad \begin{cases} r_1 = -\beta + \sqrt{\beta^2 - \omega_0^2} \\ r_2 = -\beta - \sqrt{\beta^2 - \omega_0^2} \end{cases},$$

where $2\beta \equiv b/m$, $\omega_0 \equiv \sqrt{k/m}$.

1. Fill in the following table as you work through this activity:

damping	Condition on β	Decay parameter
none	$\beta = 0$	none
under	$\omega_0 < \beta$	β
critical	$\omega_0 = \beta$	β
over	$\omega_0 > \beta$	$\beta + \sqrt{\beta^2 - \omega_0^2}$

2. **Underdamped Oscillations (weak damping)**

   (a) For what values of r does this system still exhibit oscillatory behavior? These are underdamped solutions

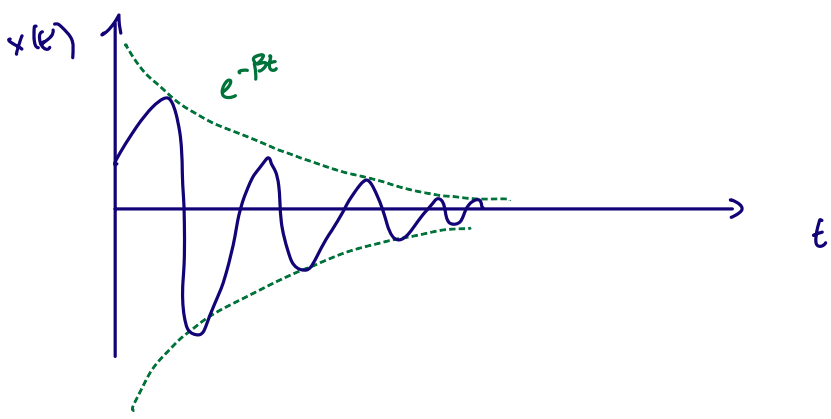
   $r = \beta \pm \sqrt{\beta^2 - \omega_0^2}$ when $\omega_0 < \beta$. Let $\omega_1^2 \equiv \omega_0^2 - \beta^2$

   $\Rightarrow r = \beta \pm i\omega_1$

   (b) Pull out an overall exponential decay factor from the general underdamped solution.

   $$x(t) = C_1 e^{(-\beta + i\omega_1)t} + C_2 e^{(-\beta - i\omega_1)t} = e^{-\beta t}\left(C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t}\right)$$

   (c) Sketch $x(t)$ for underdamped harmonic oscillator. Also sketch the envelope of the oscillations as they decay.

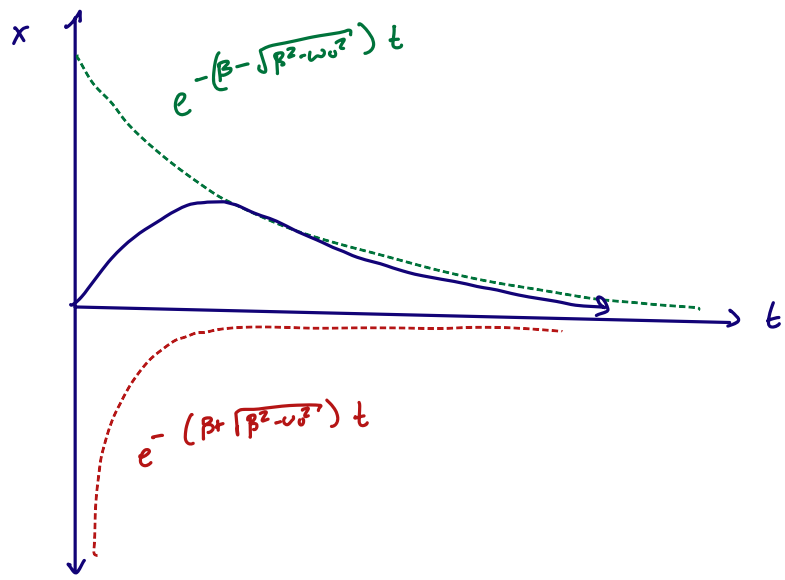


3. **Overdamping (strong damping)**

(a) For what values of r does this system **not** exhibit oscillatory behavior? These are overdamped solutions

$\beta \pm \sqrt{\beta^2 - \omega_0^2}$ where $\beta > \omega_0$

(b) Sketch the behavior of the solution of the overdamped harmonic oscillator. Assume the oscillator starts with $x_0 = 0, v_0 > 0$.



4. **Critical damping**

(a) What value of r lies at the boundary between the overdamped and underdamped scenarios? This is critical damping behavior.

$r = -\beta$, $\beta = \omega_0$

(b) How many linearly independent solutions did you find for the critical damping scenario?

one

(c) Check whether $x(t) = te^{rt}$ is a solution to the EOM in this case.

$x = te^{rt}$

$\dot{x} = e^{rt} + rte^{rt}$

$\ddot{x} = re^{rt} + re^{rt} + r^2te^{rt} = 2re^{rt} + r^2te^{rt}$

$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$

$2re^{rt} + r^2te^{rt} + 2\beta e^{rt} + 2\beta rte^{rt} + \omega_0^2 te^{rt} = 0$

$(2r + 2\beta)e^{rt} + (r^2 + 2\beta r + \omega_0^2)te^{rt} = 0$

$\Rightarrow 2r + 2\beta = 0$ and $r^2 + 2\beta r + \omega_0^2 = 0$

$\Rightarrow \boxed{r = -\beta}$ $\Rightarrow \beta^2 + 2\beta(-\beta) + \omega_0^2 = 0$

$\beta^2 - 2\beta^2 + \omega_0^2 = 0$

$-\beta^2 + \omega_0^2 = 0$

we already know $\omega_0 = \beta$

$\Rightarrow 0 = 0$ ✓

so $\boxed{te^{rt} \text{ is a solution when } r = -\beta}$

(d) Write down the general solution to the critically damped oscillator.

$x(t) = c_1 e^{-\beta t} + c_2 te^{-\beta t}$

5. Bonus: Show that, in the case of underdamping, β is the decay parameter. To get the decay parameter, find the amplitude of oscillations as a function of time. The decay parameter satisfies:

$$A(1/\beta) = \frac{A(t=0)}{e} \quad \text{typo!} \tag{2}$$

Should be: $A\left(\frac{1}{\beta}\right) = \frac{1}{e} A(t=0)$

Answer on next page.

$$x(t) = e^{-\beta t}\left(C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t}\right) \qquad \omega_1 \equiv \sqrt{\omega_0^2 - \beta^2}$$

$$= e^{-\beta t}\left(C_1 \cos\omega_1 t + i C_1 \sin\omega_1 t + C_2 \cos\omega_1 t - i C_2 \sin\omega_1 t\right)$$

$$= e^{-\beta t}\left(B_1 \cos\omega_1 t + B_2 \sin\omega_1 t\right), \qquad B_1 = C_1 + C_2$$

$$B_2 = i(C_1 - C_2)$$

$$= e^{-\beta t} A_0 \cos(\omega_1 t - \delta)$$

where $A_0^2 = B_1^2 + B_2^2$, $\delta = \tan^{-1}\left(\frac{B_1}{B_2}\right)$ } This comes from turning off $e^{-\beta t}$ & examining only the oscillatory part of the solution. Then check lecture notes

If $x(t) = A(t)\cos(\omega_1 t - \delta)$

where $A(t) = A_0 e^{-\beta t}$

$A(0) = A_0$

let decay parameter be $\bar{\beta}$:

$$A(1/\bar{\beta}) = A(t = 1/\bar{\beta}) = A_0 e^{-\beta/\bar{\beta}}$$

Also $A(1/\bar{\beta}) = \frac{1}{e} A(0) = \frac{1}{e} A_0 e^0 = \frac{1}{e} A_0$

So $\cancel{A_0} e^{-\beta/\bar{\beta}} = \frac{1}{e}\cancel{A_0}$

$$e^{-\beta/\bar{\beta}} = e^{-1}$$

$$-\beta/\bar{\beta} = -1$$

$\boxed{\beta = \bar{\beta}}$ $\Rightarrow$ decay parameter $= \beta$